

	AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
BURIED	822 A	187.89 MVA	90.0 degC	78.7 degC	34.922 W/m	OK
DUCT	683 A	156.09 MVA	90.0 degC	82.4 degC	33.404 W/m	OK
FREE AIR	991 A	226.56 MVA	90.0 degC	73.6 degC	50.770 W/m	OK

1. Design Data & Input Parameters

Parameter	Symbol / Formula	Value	Unit
Conductor material		Cu	
Conductor size		630	mm2
Conductor type		stranded_compact	
Conductor diameter Dc		30.3	mm
Conductor screen OD Dc_sc		33.3	mm
Max conductor temp		90	degC
Insulation material		XLPE	
Insulation outer diameter Di		64.3	mm
Insulation screen OD Di_sc		66.9	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
Screen type		Al	
Screen mean diameter ds		67.7	mm
Screen area As		170.2	mm2
Oversheath material		PE	
Cable outer diameter De		75.5	mm
Rated voltage U		132.0	kV
U0 line-to-earth		76.210	kV
Frequency		50	Hz
Bonding		solid	
Formation		trefoil	
Depth L (buried)		1000	mm
Spacing s (buried)		75.5	mm
Ambient ground temp		20	degC
Soil resistivity rho_soil		1.00	K.m/W
Duct installation		Do=140.0mm, Di=119.4mm, mat=HDPE, touching	
Ambient air temp		25	degC
Air mounting		free	
Solar radiation		H=1000 W/m2, sigma=0.40	

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Symbol / Formula	Value	Unit
IEC 60228 R0 at 20 degC		2.8300e-05	ohm/m
Temperature coefficient a20		3.9300e-03	1/K
R0 at theta_amb = 20 degC		2.8300e-05	ohm/m
R0 at theta_max = 90 degC		3.6085e-05	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Symbol / Formula	Value	Unit
ks strand constant (IEC 60287-1-1 Table 2)		1.0000	
kp proximity constant (IEC 60287-1-1 Table 2)		1.0000	
$xs^2 = ks \cdot (8\pi f / R0) \cdot 1e-7$		3.4824042386	
$xs_{\text{■}} = (xs^2)^2$		12.1271392811	
ys skin effect factor		0.0601241268	
$xp_{\text{■}} = (xp^2)^2$		12.1271392811	
yp" intermediate proximity (s=spacing_s)		0.0601241268	
yp proximity factor (s=spacing_s)		0.0351000648	
R at theta_max AC (buried/air)		3.9521526380e-05	ohm/m
yp proximity factor (s=Do duct)		0.0101077563	
R at theta_max AC (duct)		3.8619670681e-05	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Symbol / Formula	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
U0 line-to-earth		76.2100	kV
Di insulation outer diameter		64.3000	mm
Dc_sc conductor screen OD		33.3000	mm
Capacitance C = $er / (18 \ln(Di/Dc_sc)) \cdot 10^{-9}$		0.211077	nF/m
omega = 2 pi f		314.159265	rad/s
Wd = omega C U0^2 tan(delta)		3.8513583657e-01	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Symbol / Formula	Value	Unit
Screen material		Al	
Screen resistivity rho_s		2.8400e-08	ohm.m
Screen area As		170.1500	mm2
Wire/sheath outer OD D_sc_outer		68.5000	mm
Lay factor LFs		1.000000	
Screen Rs0 at 20 degC		1.6691e-04	ohm/m
X_sheath (s=spacing_s buried)		5.0403e-05	ohm/m
X_sheath_duct (s=Do duct)		8.9203e-05	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Symbol / Formula	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Symbol / Formula	Value	Unit
rho_ins insulation		3.50	K.m/W
rho_sc semiconducting screens		2.50	K.m/W
rho_tape bedding tape under wire screen		6.00	K.m/W
T1_cs conductor screen		0.0375644199	K.m/W
T1_i insulation		0.3665350785	K.m/W
T1_si insulation screen		0.0157719906	K.m/W
T1_uwbt bedding tape under wire screen		0.0000000000	K.m/W
T1_raw = sum of all layers		0.4198714890	K.m/W
T1 correction x1.00 (foil=False, cover=100.0%)		0.4198714890	K.m/W
T2 buried (screen to foil tape)		0.0000000000	K.m/W
T3 reference diameter D_start		68.5000	mm
T3_raw oversheath (single-layer)		0.0541996092	K.m/W
T3 buried x1.6 (touching trefoil factor)		0.0867193748	K.m/W
T3 duct x1.0 (cables not touching in duct)		0.0541996092	K.m/W
T3 air x1.0 (no touching factor in air)		0.0541996092	K.m/W
u = 2L/De (buried)		26.490066	
T4 buried (soil)		1.5945226972	K.m/W
T4 duct (iterative theta_m — see duct section)		see duct section	
T4 air (iterative x — see air section)		see air section	

8. Electrical Parameters

Parameter	Symbol / Formula	Value	Unit
Inductance L		0.160439	uH/m
Reactance X = omega L		50.4033	uOhm/m
Pos-seq resistance R1		51.1372	uOhm/m
Pos-seq impedance Z1		71.8019	uOhm/m
Capacitance C		0.211077	nF/m
Charging current Ic		5.0536	mA/m

9. Sheath Losses at Convergence (Buried)

Parameter	Symbol / Formula	Value	Unit
theta_s at convergence		78.712031	degC
Rs at theta_s		2.0640e-04	ohm/m
X sheath reactance		5.0403e-05	ohm/m
lam1' circulating		0.2939073180	
lam1" eddy		0.0000000000	
lam1 = lam1' + lam1"		0.2939073180	
lam2 armour		0.000000	

10. Ampacity — Direct Buried [IEC 60287-1-1 Cl.1.4.1.1]

Parameter	Symbol / Formula	Value	Unit
Delta_theta		70.00	K
Etop numerator		69.2716396495	
Ebot denominator		1.0257e-04	
I = sqrt(Etop/Ebot) exact		821.8108532786	A
I rounded TB880 GP2		822	A
MVA = sqrt(3) x U x I		187.8912	MVA
Iterations		5	
Convergence		YES	

11. Temperature Profile — Buried

Parameter	Symbol / Formula	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.000000	degC
theta_s sheath surface		78.712031	degC
theta_j cable surface		75.683633	degC
theta_a ambient		20.0	degC

12. Loss Budget — Buried

Parameter	Symbol / Formula	Value	Unit
Conductor Wc		26.691775	W/m
Sheath Ws		7.844908	W/m
Armour Wa		0.000000	W/m
Dielectric Wd		3.851358e-01	W/m
Total		34.921819	W/m

13. Duct Installation Parameters [IEC 60287-2-1 Cl.4.2.7]

Parameter	Symbol / Formula	Value	Unit
Duct material		HDPE	
Formation		trefoil	
Touching ducts		Yes	
Duct outer diameter Do		140.0	mm
Duct inner diameter Di		119.4	mm
Di/De		1.581	
U (IEC Table 4)		1.87	
V (IEC Table 4)		0.312	
Y (IEC Table 4)		0.0037	
theta_m initial		70.0	degC

14. Duct T4 Components [IEC 60287-2-1 Cl.4.2.7]

Parameter	Symbol / Formula	Value	Unit
T4' air gap (converged)		0.3433990137	K.m/W
T4" duct wall		0.0886606472	K.m/W
T4''' soil external		1.3806035704	K.m/W
T4 total		1.8126632312	K.m/W
theta_m converged		74.8151	degC

15. Sheath Losses at Convergence (Duct)

Parameter	Symbol / Formula	Value	Unit
theta_s at convergence		82.361097	degC
Rs at theta_s		2.0886e-04	ohm/m
X sheath reactance (s=Do)		8.9203e-05	ohm/m
lam1' circulating		0.8343056836	
lam1" eddy		0.0000000000	
lam1 = lam1' + lam1"		0.8343056836	
lam2 armour		0.000000	

16. Ampacity — Duct [IEC 60287-1-1 Cl.1.4.1.1]

Parameter	Symbol / Formula	Value	Unit
Delta_theta		70.00	K
Etop numerator		69.2001504396	
Ebot denominator		1.4846e-04	
I = sqrt(Etop/Ebot) exact		682.7196040417	A
I rounded TB880 GP2		683	A
MVA = sqrt(3) x U x I		156.0907	MVA
Iterations		6	
Convergence		YES	

17. Temperature Profile — Duct

Parameter	Symbol / Formula	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.000000	degC
theta_s sheath surface		82.361097	degC
theta_j cable surface		80.550602	degC
theta_duct_inner		69.079625	degC
theta_m mean air gap		74.8151	degC
theta_a ambient		20.0	degC

18. Loss Budget — Duct

Parameter	Symbol / Formula	Value	Unit
Conductor Wc		18.000862	W/m
Sheath Ws		15.018222	W/m
Armour Wa		0.000000	W/m
Dielectric Wd		3.851358e-01	W/m
Total		33.404220	W/m

19. Free Air Parameters [IEC 60287-2-1 Cl.2.2.1]

Parameter	Symbol / Formula	Value	Unit
Formation		trefoil	
Mounting		free	
Z (IEC Table 2)		0.9600	
E (IEC Table 2)		1.2500	
g (IEC Table 2)		0.2000	
De* (m)		0.075500	m
$h = Z/(De^*)^g + E$		2.8594661337	W/(m ² K ^{5/4})
Solar		Yes	
H		1000.0	W/m ²
sigma		0.40	
theta_a_air		25.0	degC
Delta_theta		65.00	K

20. Free Air T4 [IEC 60287-2-1 Cl.2.2.1]

Parameter	Symbol / Formula	Value	Unit
$x = (\theta_j - \theta_a)^{1/4}$		2.6024876298	K ^(1/4)
$D\theta_s = \theta_j - \theta_a$		45.8727414814	K
$T4 = 1/(\pi \cdot De \cdot h \cdot x)$		0.5665386515	K.m/W
Solar $\sigma \cdot De \cdot H \cdot T4$		17.1094672748	K

21. Sheath Losses at Convergence (Air)

Parameter	Symbol / Formula	Value	Unit
theta_s at convergence		73.624466	degC
Rs at theta_s		2.0298e-04	ohm/m
X sheath reactance		5.0403e-05	ohm/m
lam1'		0.2982916317	
lam1''		0.0000000000	
lam1		0.2982916317	
lam2		0.000000	

22. Ampacity — Free Air [IEC 60287-1-1 Cl.1.4.4.1]

Parameter	Symbol / Formula	Value	Unit
Etop numerator (incl solar)		47.5706103973	
Ebot denominator		4.8444e-05	
$I = \sqrt{Etop/Ebot}$ exact		990.9411775310	A
I rounded TB880 GP2		991	A
$MVA = \sqrt{3} \times U \times I$		226.5596	MVA
Iterations		5	
Convergence		YES	

23. Temperature Profile — Free Air

Parameter	Symbol / Formula	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.000000	degC
theta_s sheath surface		73.624466	degC
theta_j cable surface		70.872741	degC
theta_a ambient air		25.0	degC

24. Loss Budget — Free Air

Parameter	Symbol / Formula	Value	Unit
Conductor Wc		38.808733	W/m
Sheath Ws		11.576320	W/m
Armour Wa		0.000000	W/m
Dielectric Wd		3.851358e-01	W/m
Solar heat		17.109467	W/m
Total		50.770189	W/m

25. Electrical Parameters

Parameter	Symbol / Formula	Value	Unit
Inductance L		0.160439	uH/m
Reactance X = omega L		50.4033	uOhm/m
Pos-seq resistance R1		51.1372	uOhm/m
Pos-seq impedance Z1		71.8019	uOhm/m
Capacitance C		0.211077	nF/m
Charging current Ic		5.0536	mA/m

Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880.

TB880 options applied — GP2: rounded down. GP6: eddy losses. GP7: dielectric losses. GP8: exact T4. T3: 1.6x touching trefoil (buried only).

Solar radiation included: H=1000 W/m2, sigma=0.40.

Duct: HDPE Do=140.0mm Di=119.4mm touching trefoil.

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